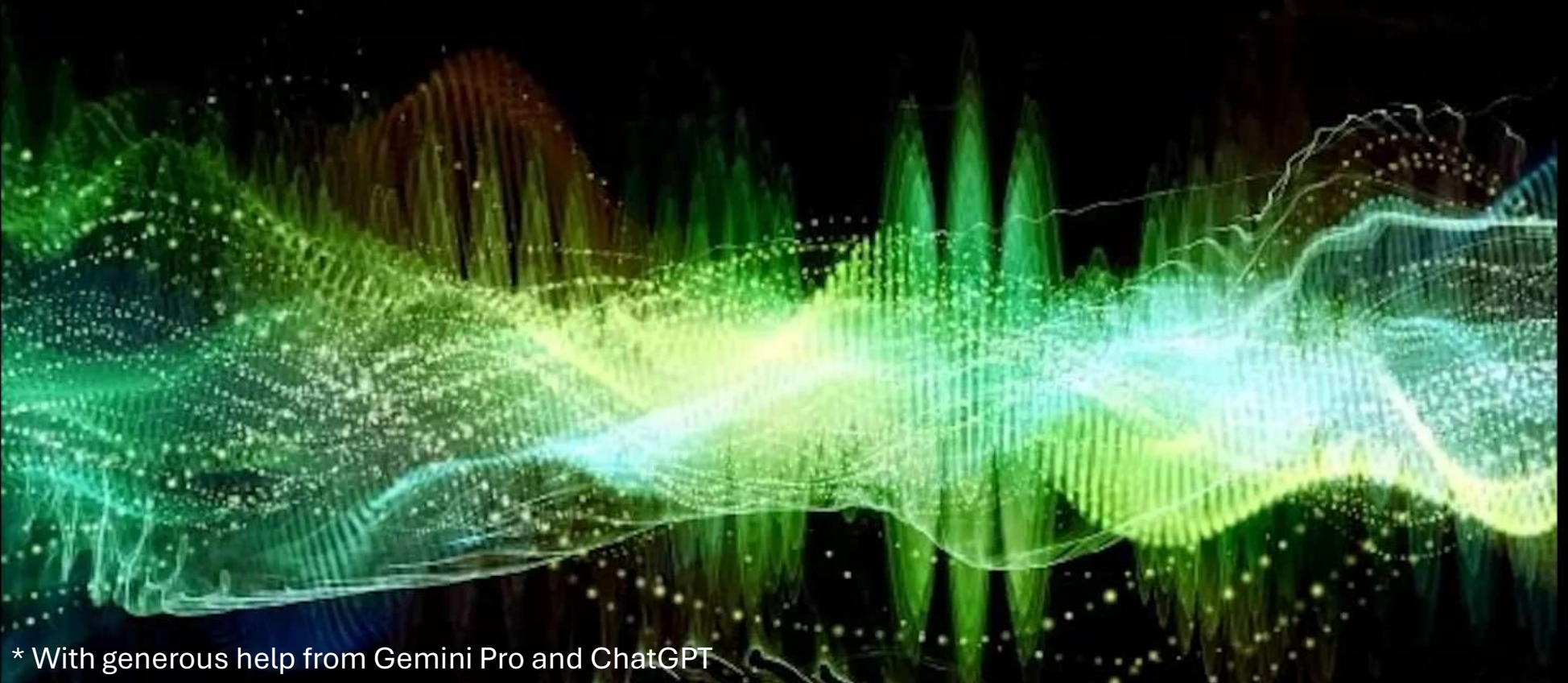


# **ECEN 4/5632 Theory and Application of Digital Filters**

**Lecture 4, 9-03-26**

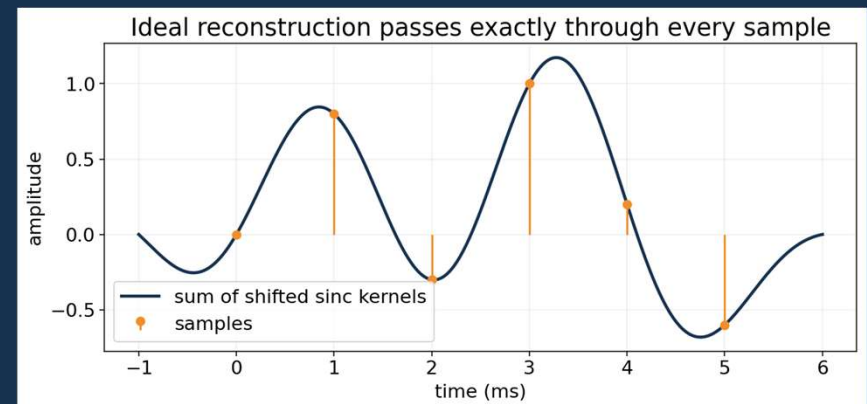
**Peter Mathys\***



\* With generous help from Gemini Pro and ChatGPT

# Reconstruction, D/A, and Quantization Noise

Ideal interpolation gives the theory; ZOH and finite bit depth explain what hardware actually does.



75 minutes • ideal D/C • sinc interpolation • zero-order hold • quantization • SQNR • PyTorch dtype bridge

## Recap: after Lecture 3, we trust the sample times

Now we ask how to reconstruct between samples and represent amplitudes.

**Sampling error**

Where Week 2 lives in the audio → ML pipeline

**Quantization error**



Aliasing moves energy to the wrong frequency.  
It is fundamentally non-unique.

Quantization maps a continuous amplitude to  
one of finitely many levels.

# Learning outcomes

By the end of Lecture 4, students should be able to move between ideal theory and real hardware.

- Derive the ideal bandlimited interpolation formula from the reconstruction low-pass filter.
- Explain zero-order hold behavior in time and its sinc-shaped magnitude droop.
- Model a uniform B-bit quantizer and compute  $\Delta$  and quantization-error variance.
- Derive and use  $\text{SQNR} \approx 6.02B + 1.76$  dB for a full-scale sine.
- Explain the difference between 16-bit PCM storage and float32 computation in PyTorch.

# Ideal reconstruction in the frequency domain

If spectral replicas do not overlap, simply isolate the baseband copy.

$$X_s(j\Omega) = \frac{1}{T} \sum_k X_c(j(\Omega - k\Omega_s))$$

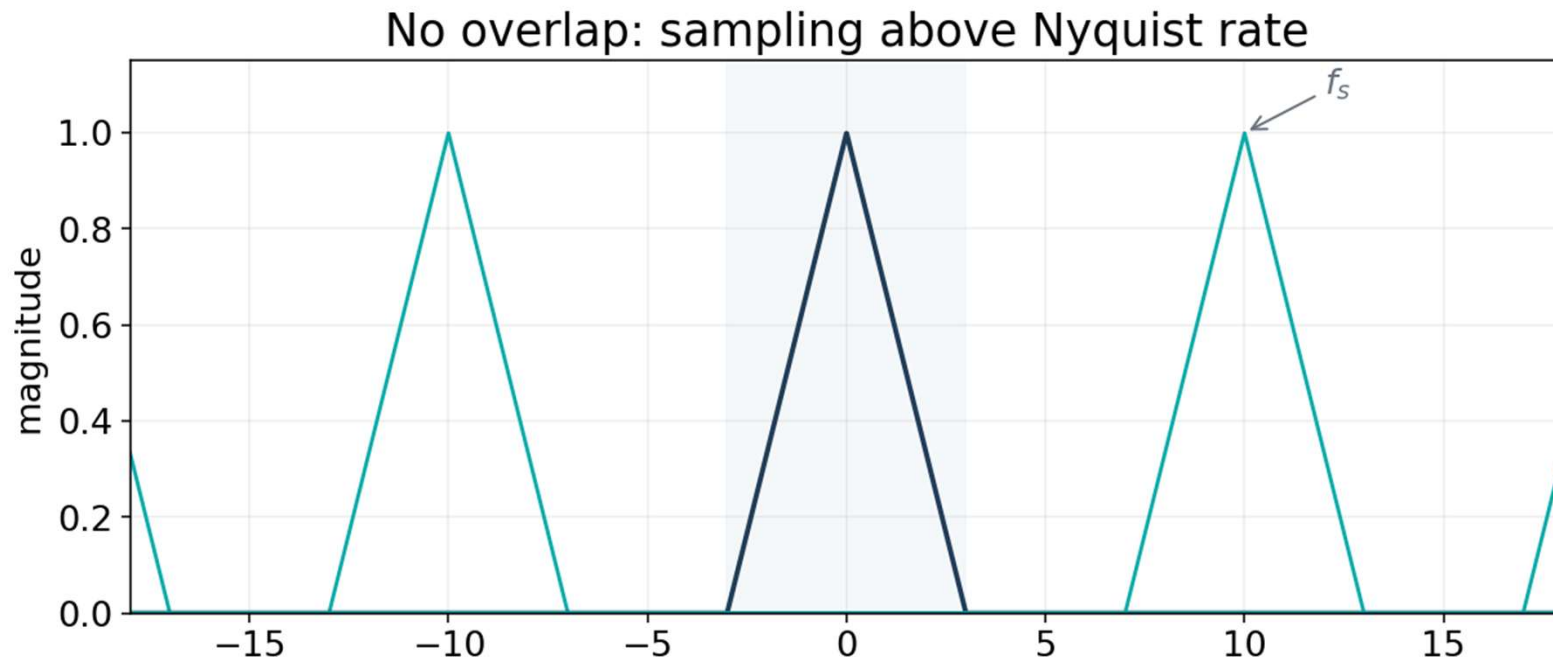
$$H_r(j\Omega) = \begin{cases} T, & |\Omega| < \Omega_c \\ 0, & \text{otherwise} \end{cases}$$

**The factor T cancels the 1/T introduced by ideal sampling.**

Choose  $\Omega_M < \Omega_c < \Omega_s - \Omega_M$  so only one replica passes.

# What the sampled spectrum looks like

Every copy contains the same analog information—until the copies overlap.



Replica spacing =  $f_s$ . Baseband can be isolated only when neighboring replicas do not overlap.

# From ideal low-pass filtering to sinc interpolation

The inverse transform of a brick-wall low-pass filter is a sinc.

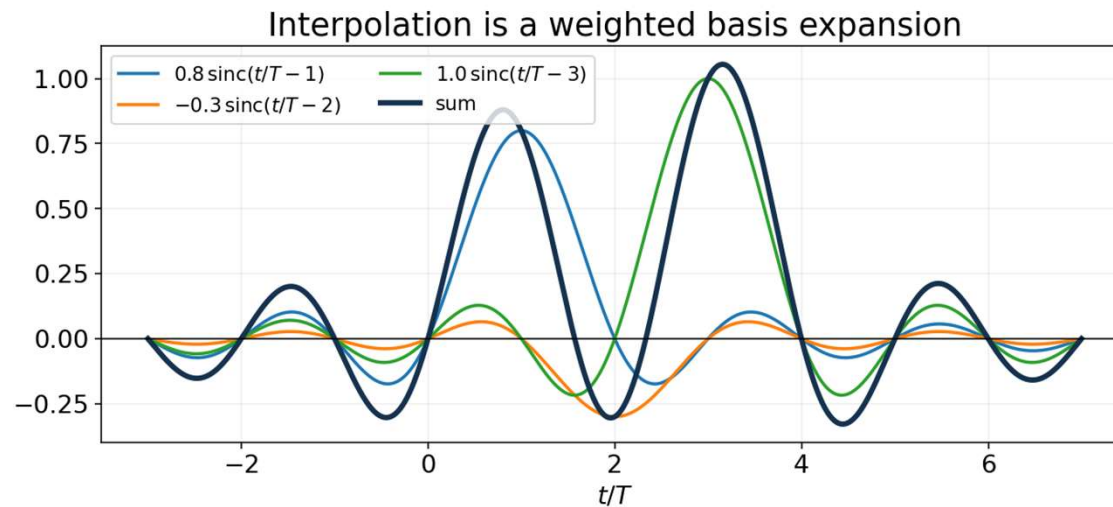
$$h_r(t) = \text{sinc}\left(\frac{t}{T}\right) \quad \text{for the ideal Nyquist-band reconstruction case}$$

$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n] \text{sinc}\left(\frac{t - nT}{T}\right)$$

**Here  $\text{sinc}(u) = \sin(\pi u)/(\pi u)$ . Every sample contributes one shifted, scaled interpolation kernel.**

# Why the sinc basis passes through every sample

At integer offsets, normalized sinc is zero—except at the center.

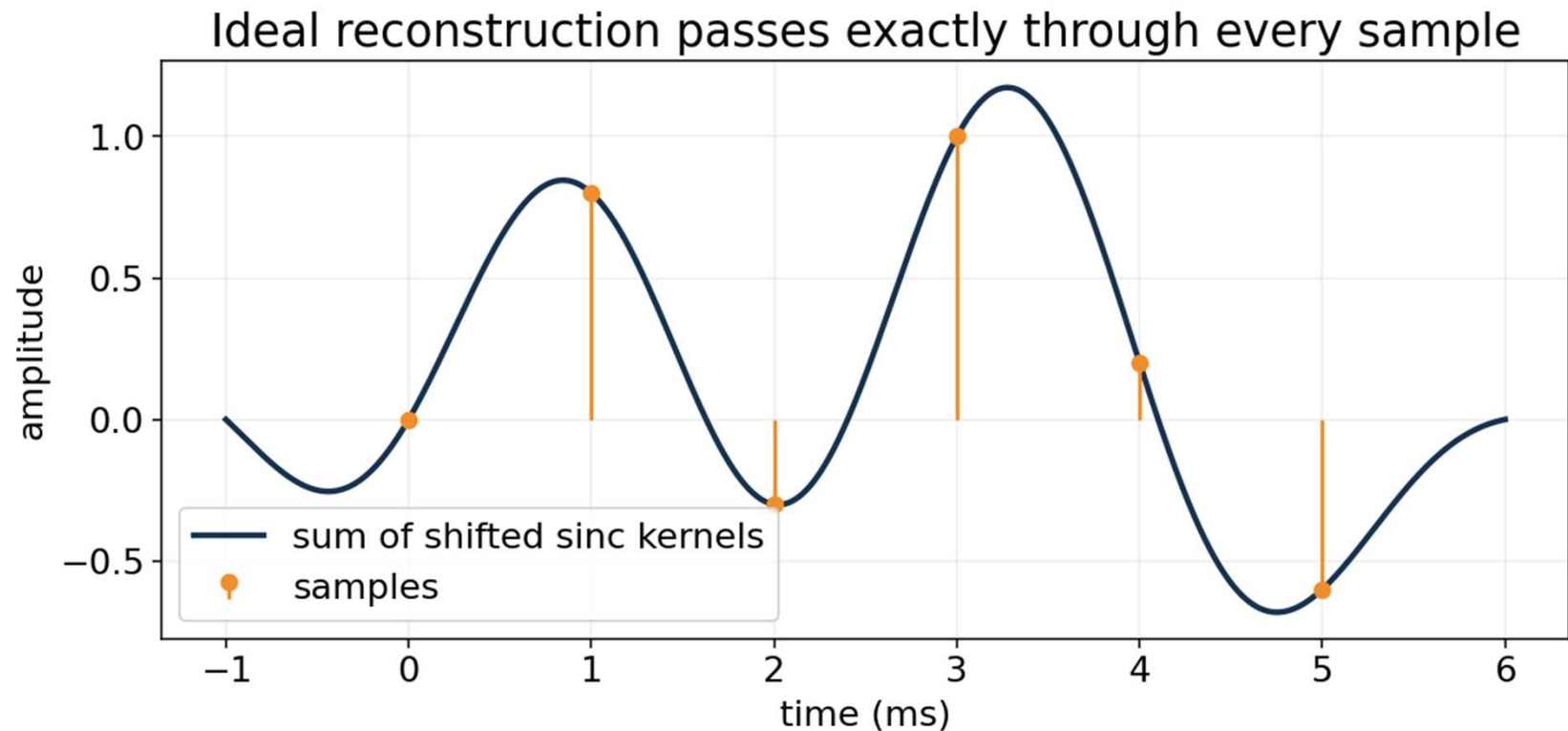


At  $t=mT$ , only the  $n=m$  kernel survives  $\rightarrow x_r(mT)=x[m]$ .



# A finite illustration of ideal interpolation

The true ideal sum is infinite; a finite example still shows the geometry.



# Why a real DAC does not implement ideal sinc directly

The ideal impulse response is infinite in duration and noncausal when centered at  $t=0$ .

## Ideal mathematical interpolator

- Infinite support in time.
- Symmetric around the interpolation instant.
- Brick-wall frequency response.
- Perfect only under ideal bandlimiting assumptions.

## Practical hardware strategy

- Generate a simple analog waveform from each code.
- Zero-order hold is extremely natural in a DAC.
- Follow with an analog reconstruction / anti-imaging filter.
- Oversampling reduces the burden on analog filtering.

**Ideal theory tells us the target; implementation approximates it.**

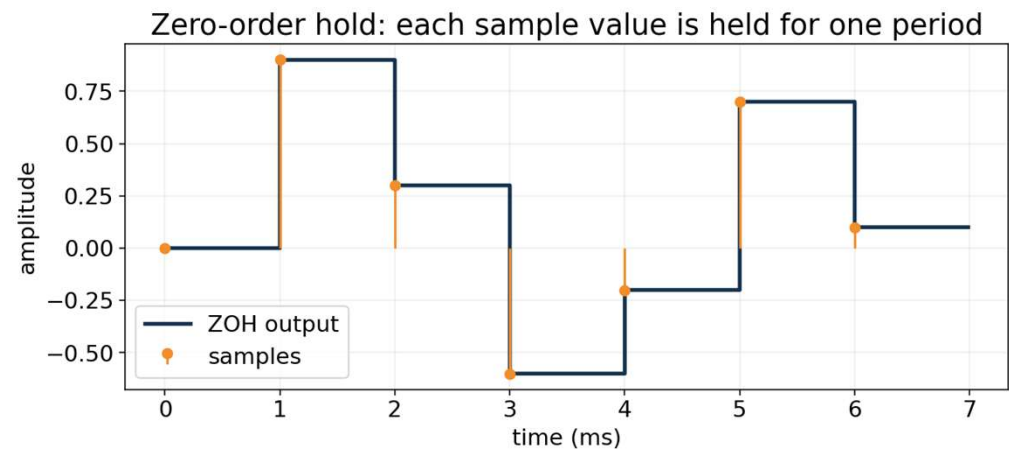
# Zero-order hold (ZOH): hold each sample for one period

A rectangular pulse is convolved with the weighted impulse train.

$$h_{ZOH}(t) = u(t) - u(t - T)$$

$$x_{ZOH}(t) = x_s(t) * h_{ZOH}(t)$$

This staircase is already a continuous-time waveform—but it contains spectral images and amplitude droop.



# Deriving the ZOH frequency response

Transform the rectangular hold pulse.

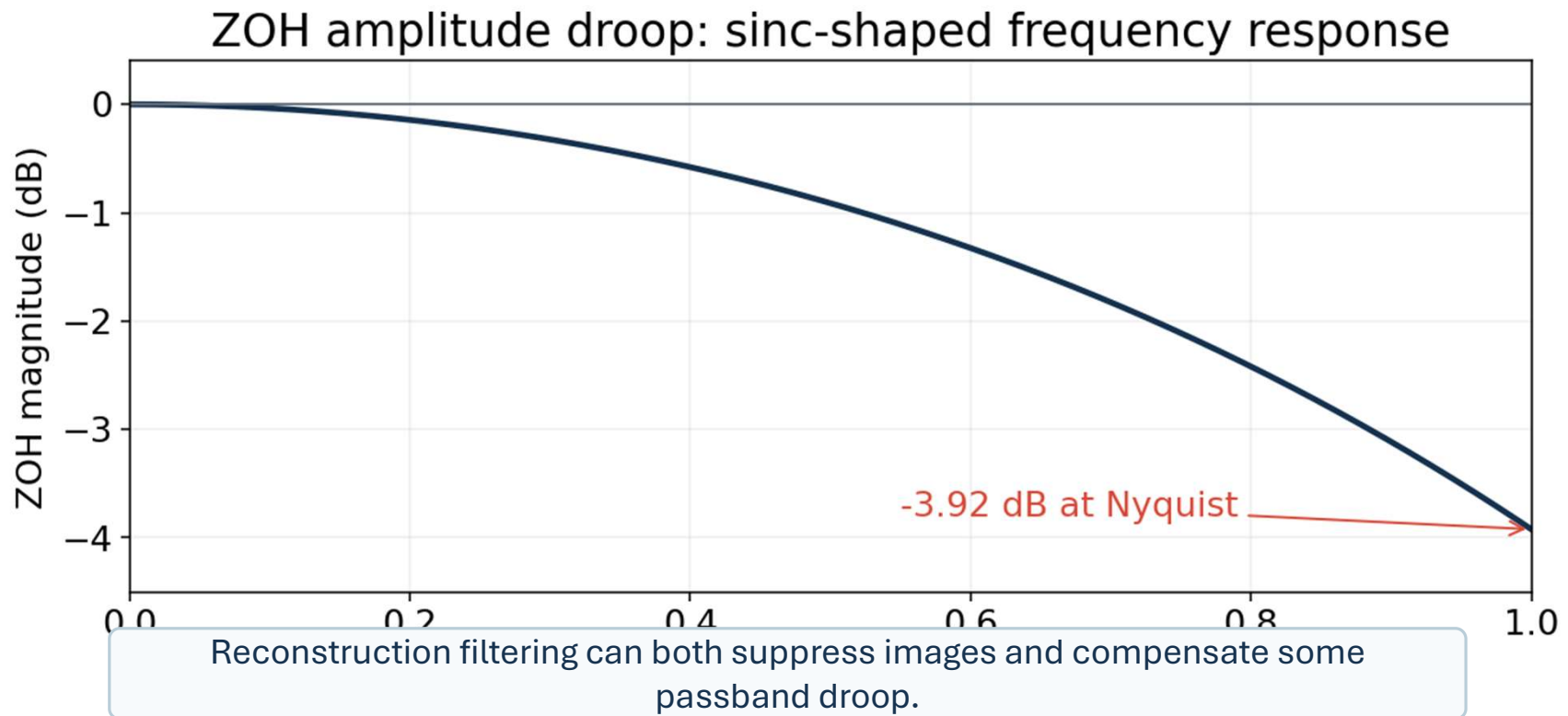
$$H_{ZOH}(j\Omega) = \int_0^T e^{-j\Omega t} dt$$
$$= T e^{-j\Omega T/2} \frac{\sin(\Omega T/2)}{\Omega T/2}$$

- Linear phase term:  $e^{-j\Omega T/2}$  corresponds to a  $T/2$  delay.
- Magnitude term: sinc-shaped envelope.
- High frequencies are attenuated more than low frequencies.

**At  $\Omega T = \pi$ :  $|H|/T = 2/\pi \approx 0.637 \rightarrow -3.92$  dB.**

# The ZOH creates sinc-shaped amplitude droop

This is deterministic frequency response—not random noise.



# A practical D/A chain

ZOH is only one stage; analog filtering completes the reconstruction.



The analog LPF removes spectral images around multiples of  $f_s$ ; oversampling gives the filter more transition width.

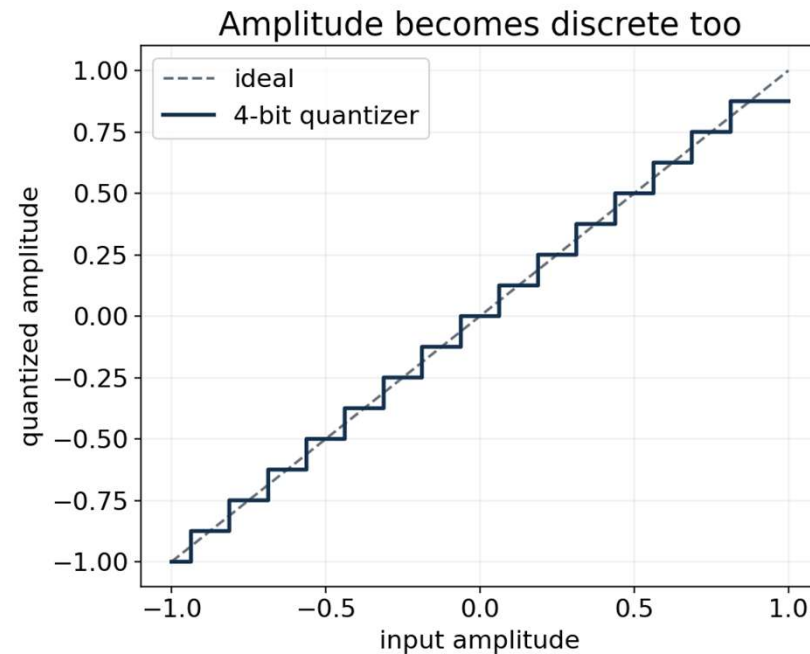
# Quantization: the second discretization

Sampling discretizes time. Quantization discretizes amplitude.

$$x_q[n] = Q\{x[n]\} = x[n] + e[n]$$

$$\Delta = \frac{2A}{2^B}$$

- B bits  $\rightarrow 2^B$  nominal levels.
- $\Delta$  is the quantizer step for full-scale range roughly  $[-A, A]$ .
- Overload/clipping is a separate failure mode when  $|x|$  exceeds the designed range.



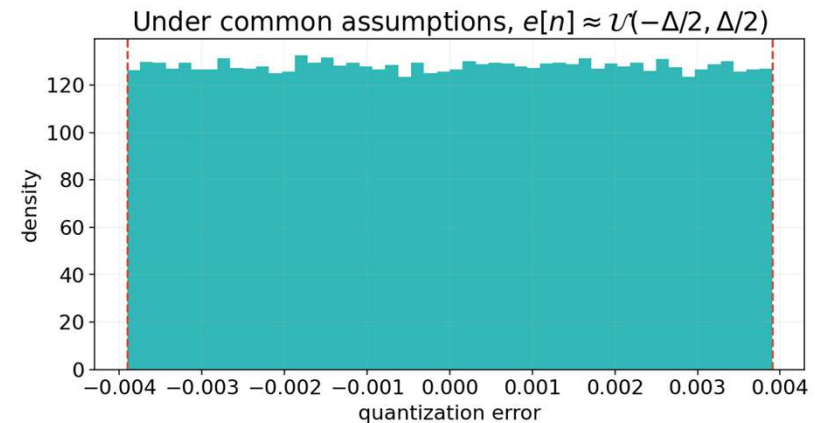
# The additive quantization-noise model

Useful—but only under assumptions.

$$e[n] = x_q[n] - x[n]$$

$$e[n] \approx \mathcal{U}(-\Delta/2, \Delta/2)$$

$$\mathbb{E}\{e[n]\} \approx 0, \quad \sigma_e^2 \approx \frac{\Delta^2}{12}$$



Good approximation when the signal spans many levels and the error is sufficiently decorrelated from the signal.



## Derive the quantization-noise variance

Uniform error over one quantization interval.

$$p_e(e) = \frac{1}{\Delta}, \quad -\Delta/2 \leq e < \Delta/2$$

$$\sigma_e^2 = \int_{-\Delta/2}^{\Delta/2} e^2 \frac{1}{\Delta} de = \frac{\Delta^2}{12}$$

**Every additional bit halves  $\Delta \rightarrow$  noise power falls by a factor of 4  $\rightarrow$  about 6 dB improvement.**

## Full-scale sine SQNR: where $6.02B + 1.76$ dB comes from

This is not an arbitrary rule of thumb; it follows from signal and error power.

$$P_x = \frac{A^2}{2}, \quad P_e = \frac{\Delta^2}{12}, \quad \Delta = \frac{2A}{2^B}$$

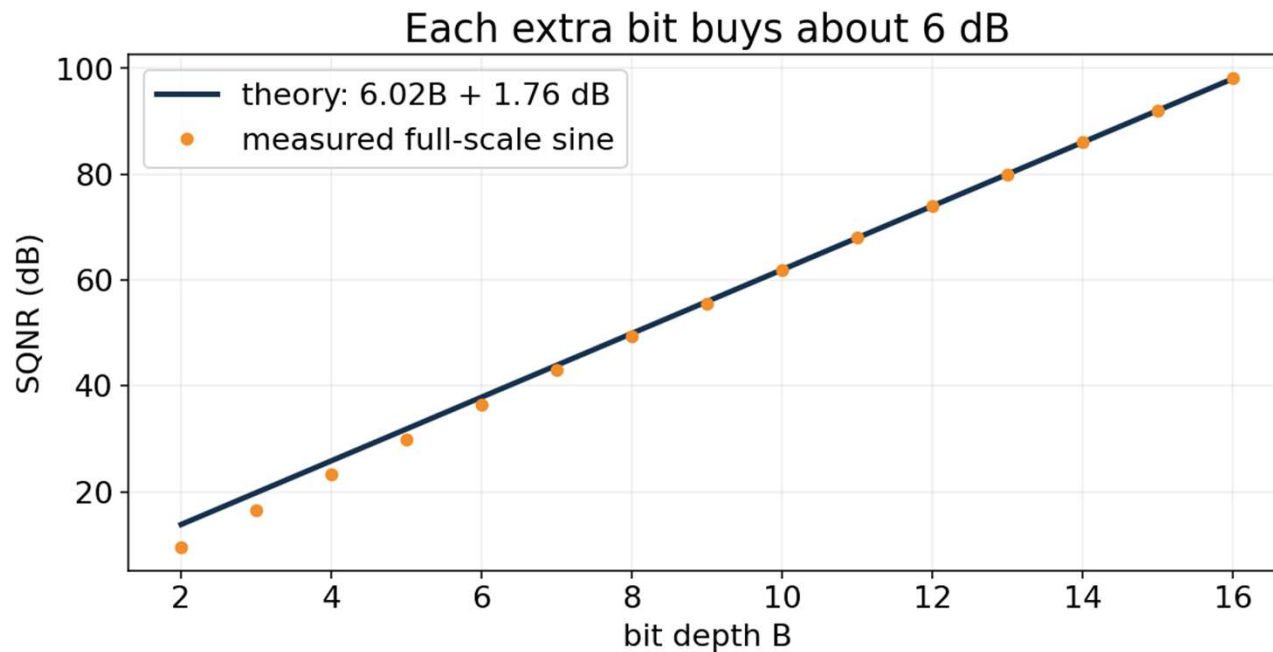
$$\text{SQNR} = 10 \log_{10} \frac{P_x}{P_e} = 10 \log_{10} \left( \frac{3}{2} 2^{2B} \right)$$

$$\text{SQNR} \approx 6.02B + 1.76 \text{ dB}$$

The +1.76 dB constant is specific to a full-scale sine and the ideal uniform-quantizer model.

# About 6 dB per bit—verified numerically

The simple model is remarkably accurate at moderate/high resolution.



**4-bit results deviate more because the high-resolution assumptions are weak.**

**16-bit full-scale sine theory: 98.08 dB SQNR.**

# 16-bit audio: “96 dB” and “98 dB” can both appear

They refer to slightly different reference calculations.

## 6 dB/bit rule of thumb

$$20 \log_{10}(2^{16}) \approx 96.33 \text{ dB}$$

96.3 dB is the amplitude ratio across 16 binary bits.

## Full-scale sine SQNR

$$6.02(16) + 1.76 \approx 98.08 \text{ dB}$$

98.1 dB includes the RMS power of a full-scale sine and uniform error model.

**Be precise about what quantity is being compared.**

# PCM storage is not the same thing as ML compute dtype

Casting to float changes representation—not the information already captured.

```
pcm = np.array(..., dtype=np.int16)
x = pcm.astype(np.float32) / 32768.0
xt = torch.from_numpy(x)
# xt.dtype == torch.float32
```

- 16-bit PCM: discrete integer codes from the recording/ADC pipeline.
- float32 tensor: convenient numerical format for DSP and neural-network arithmetic.
- Converting int16  $\rightarrow$  float32 cannot recreate amplitude values between the original quantization levels.
- A float32 tensor can still contain badly aliased or coarsely quantized data.

**More bits in the variable  $\neq$  more bits of information in the measurement.**

# Live audio demo: listen in this order

The package includes pre-generated WAV files and notebook code.

- 1 **Clean 48 kHz** reference
- 2 **Naive 48→12 kHz**
- 3 **Filtered 48→12 kHz**
- 4 **8-bit quantized**
- 5 **4-bit quantized**

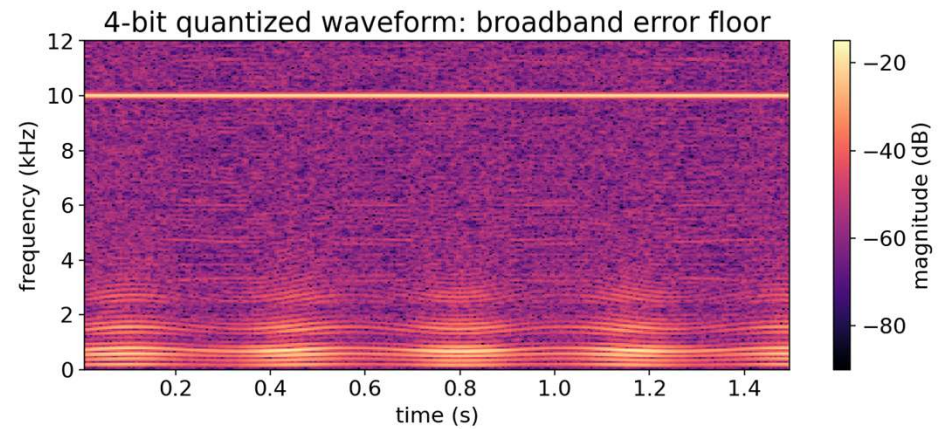
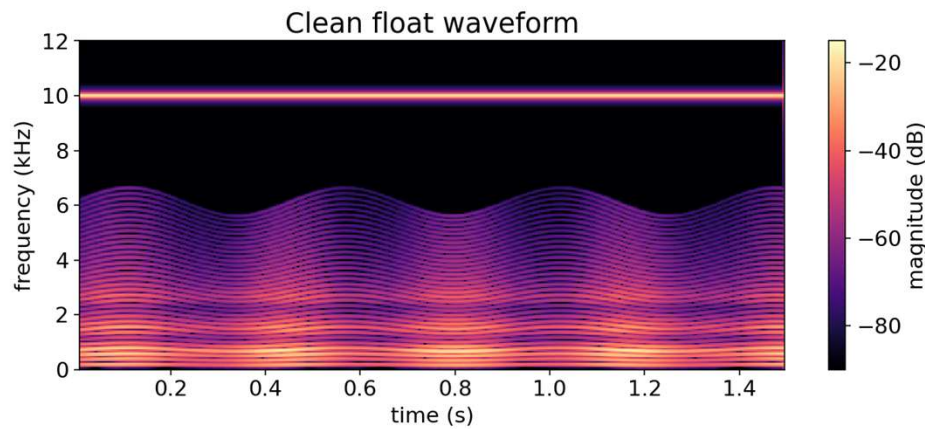
**Ask students to predict what changes before playing each file.**

# What 4-bit quantization does to a spectrogram

The error is often broad-band, but it need not be perfectly white or independent.

Clean

4-bit



# Graduate nuance: when the “white quantization noise” story breaks

Dither is one tool for trading correlated distortion for more noise-like error.

- Low-level sinusoids can produce periodic, signal-correlated error rather than white noise.
- Overload clipping is nonlinear distortion and is not captured by  $\Delta^2/12$ .
- Dither intentionally adds noise before quantization to decorrelate the quantization error.
- Noise shaping can push quantization noise toward frequency regions that matter less—but requires feedback/oversampling ideas beyond today.

**Modeling lesson for ML:  
know which “noise  
assumptions” are physics  
and which are convenient  
approximations.**



# Exit check

Students answer individually; collect or discuss in the last two minutes.

**1 Why does ideal reconstruction use sinc functions?**

**2 What deterministic frequency-domain effect does a ZOH introduce?**

**3 A full-scale 12-bit sine has approximately what ideal SQNR?**

# Lecture 4 summary: faithful digital audio needs both time and amplitude integrity

Sampling theory, reconstruction, and quantization are one end-to-end system.

- Ideal D/C isolates one spectrum replica and yields sinc interpolation.
- A zero-order hold is practical but imposes sinc-shaped passband droop and spectral images.
- A B-bit uniform quantizer has step  $\Delta \approx 2A/2^B$ ; under standard assumptions  $\sigma_e^2 \approx \Delta^2/12$ .
- A full-scale sine gives  $\text{SQNR} \approx 6.02B + 1.76$  dB.
- int16→float32 is a computational conversion; it does not undo aliasing or quantization.

**Week 2 principle:  
Bad acquisition creates  
“features” that no  
downstream ML method  
can reliably repair.**